1 Mean value of position angle change Mark II

This memo documents an updated model for estimating the change in position angle, $\Delta\Psi$, and its uncertainty. Let

$$P_k = Q_k + iU_k = L_k \exp i2\Psi_k \tag{1}$$

represent the linear polarization of the kth phase bin in one profile and

$$P'_k = Q'_k + iU'_k = L'_k \exp i2\Psi'_k \tag{2}$$

the same in the other profile.

Consider the **weighted** cross-correlation, $\bar{Z} = \bar{C} + i\bar{S}$, where $\bar{C}$ and $\bar{S}$ are the weighted mean values of the real and imaginary components of

$$z_k = c_k + is_k = P^*_k P'_k = L_k L'_k \exp i2(\Psi'_k - \Psi_k). \tag{3}$$

That is, where

$$c_k = Q_k Q'_k + U_k U'_k, \qquad s_k = Q_k U'_k - U_k Q'_k, \tag{4}$$

and

$$\begin{aligned}
\sigma^2_{c_k} &= Q'^2_k \sigma^2_{Q_k} + U'^2_k \sigma^2_{U_k} + Q^2_k \sigma^2_{Q'_k} + U^2_k \sigma^2_{U'_k} && (5)\\
\sigma^2_{s_k} &= U'^2_k \sigma^2_{Q_k} + Q'^2_k \sigma^2_{U_k} + U^2_k \sigma^2_{Q'_k} + Q^2_k \sigma^2_{U'_k} && (6)\\
\sigma_{s_k c_k} &= Q'_k U'_k (\sigma^2_{Q_k} - \sigma^2_{U_k}) + Q_k U_k (\sigma^2_{U'_k} - \sigma^2_{Q'_k}) && (7)
\end{aligned}$$

the weighted means and their variances and covariance are given by

$$\bar{C} = \sigma^2_{\bar{C}} \sum_{k=1}^{N} \frac{c_k}{\sigma^2_{c_k}} \qquad \bar{S} = \sigma^2_{\bar{S}} \sum_{k=1}^{N} \frac{s_k}{\sigma^2_{s_k}} \tag{8}$$

and

$$\sigma^2_{\bar{C}} = \left[\sum_{k=1}^{N} \frac{1}{\sigma^2_{c_k}} \right]^{-1} \qquad \sigma^2_{\bar{S}} = \left[\sum_{k=1}^{N} \frac{1}{\sigma^2_{s_k}} \right]^{-1} \qquad \sigma_{\bar{S}\bar{C}} = \sum_{k=1}^{N} \left[\frac{\sigma_{\bar{S}} \sigma_{\bar{C}}}{\sigma_{s_k} \sigma_{c_k}} \right]^2 \sigma_{s_k c_k} \tag{9}$$

The expectation value of $\Delta\Psi = \Psi'_k - \Psi_k$ is given by

$$\langle \Delta\Psi \rangle = \frac{1}{2} \tan^{-1} \left(\frac{\bar{S}}{\bar{C}} \right). \tag{10}$$

and the variance of the expectation value

$$\mathrm{var}(\langle \Delta\Psi \rangle) = \frac{\bar{C}^2 \sigma^2_{\bar{S}} + \bar{S}^2 \sigma^2_{\bar{C}} + 2\bar{S}\bar{C}\sigma_{\bar{S}\bar{C}}}{\left(\bar{C}^2 + \bar{S}^2\right)^2} \tag{11}$$